

UNIT 2

BASICS OF COMPUTER NETWORKS



Learning Objectives

By the end of this unit, students will be able to:

- Define a computer network and explain its significance in modern communication.
- Identify and describe key components of a network, including clients, servers, workstations, switches, routers and modems.
- Differentiate between various types of networks (PAN, LAN, MAN, WAN) and their real-world applications.
- Compare wired and wireless network connections based on their advantages and limitations.
- Understand the functions of key communication devices and their role in networking.
- Understand how the internet and devices communicate.

Introduction to Computer Networks

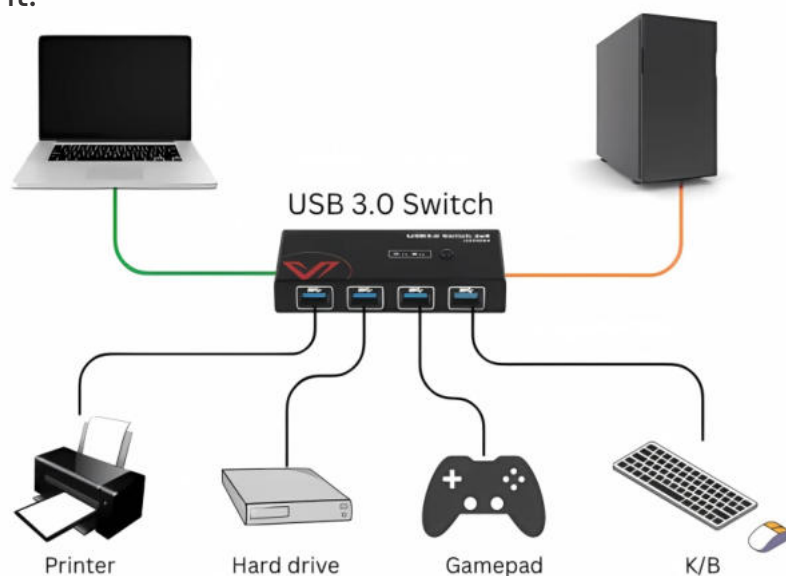
A **computer network** is like a highway for data, connecting computers and other devices to share resources and information. It enables communication between devices such as printers, scanners, and servers, making collaboration and resource sharing efficient. Networking forms the backbone of modern communication and impacts many aspects of our daily lives, from accessing the internet to sharing files in an office.

What is Networking?

Networking is the process of connecting devices (like computers, smartphones, and printers) to exchange information and share resources. It involves either wired connections (using cables) or wireless connections (using signals).

Why is Networking Important?

- 1. Communication:** Enables emails, video calls, and messaging.
- 2. Resource Sharing:** Shares files and devices like printers.
- 3. Internet Access:** Facilitates online education, research, and entertainment.



A computer network relies on various components and devices that work together to ensure smooth communication between devices. These components and devices can be categorized into two sections: network components that define the roles of devices within the network, and communication devices responsible for transmitting data.

Network Components

1. Client

A **client** is a device that requests services or resources from a server.

Example: A computer accessing a website hosted on a server.



2. Server

A **server** is a powerful computer that provides services or resources to clients.

Example: A web server hosting websites for users to access.



3. Workstation

A **workstation** is a personal computer connected to a network. It's used for tasks like browsing the internet, creating documents, or accessing shared resources.

Example: A computer in a school lab or an office workstation.



Fun Fact

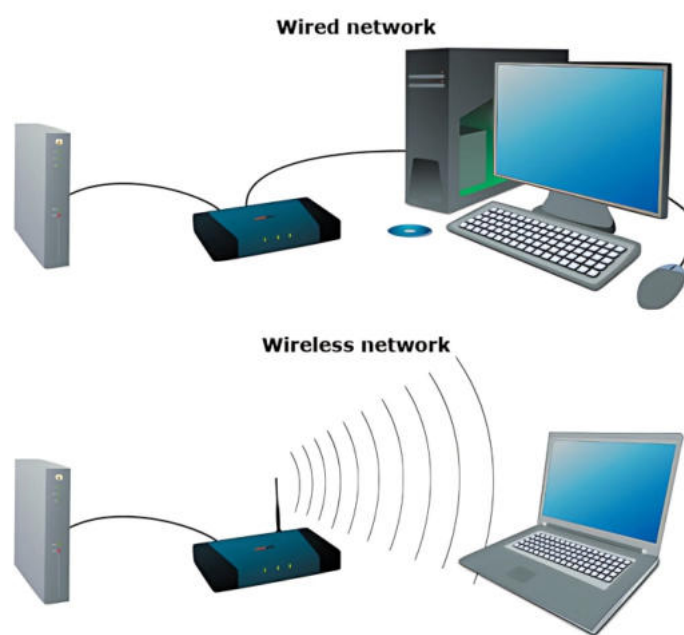
The term "server" comes from its function—it "serves" resources to clients!

Communication Devices

Communication devices enable the transmission and reception of data, instructions, and information between different devices in a network. They play a crucial role in ensuring that data reaches its destination effectively.

1. Switch

A **switch** is a device that connects multiple devices (computers, printers, scanners, etc.) within a network. It ensures that data packets are transferred to the correct destination device. Switches work at the data link layer, using MAC addresses to forward data.



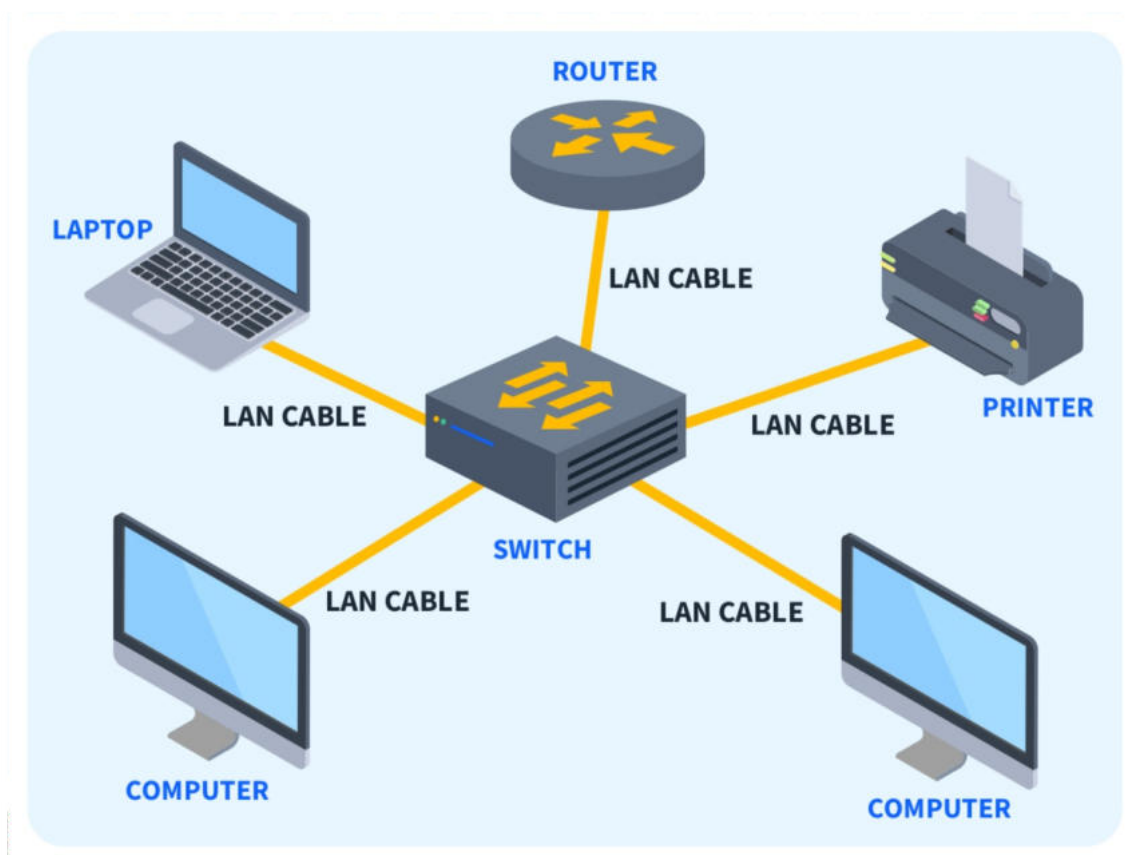
2. Router

A **router** connects different networks and directs data between them. It evaluates the best path for data to travel and manages traffic flow. Routers help in inter-network communication, such as connecting a local area network (LAN) to the internet.

Key Functions of a Router:

- **Determining the optimal data path:** Finds the best route for data to travel.

Traffic control: Manages data flow between networks.

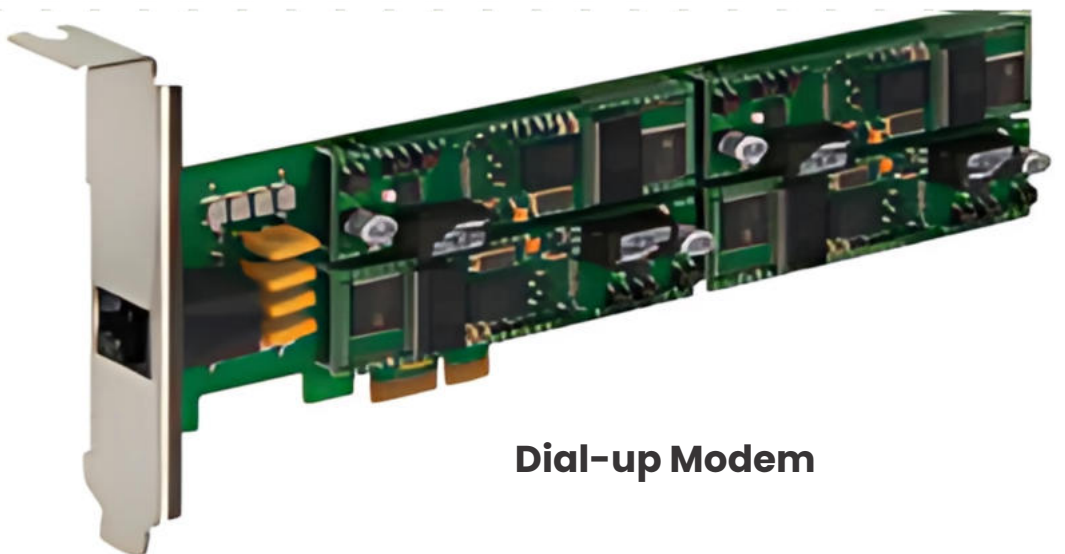


3. Modem

A **modem** (Modulator-Demodulator) is a device that converts digital signals into analog for transmission over communication lines and converts received analog signals back into digital form. Modems are essential for internet connectivity, allowing data to be transferred over phone lines, cable, or satellite connections.

Types of Modems:

- **Dial-up Modem:** It uses standard telephone lines to transmit and receive information. It converts digital signals to analog and vice versa. A dialup modem can be internal or external. It is important to remember that telephone lines carry only analog signals, whereas data packets sent by the computer are in digital form. In order to send these packets across a telephone line, modem converts digital signals into analog.



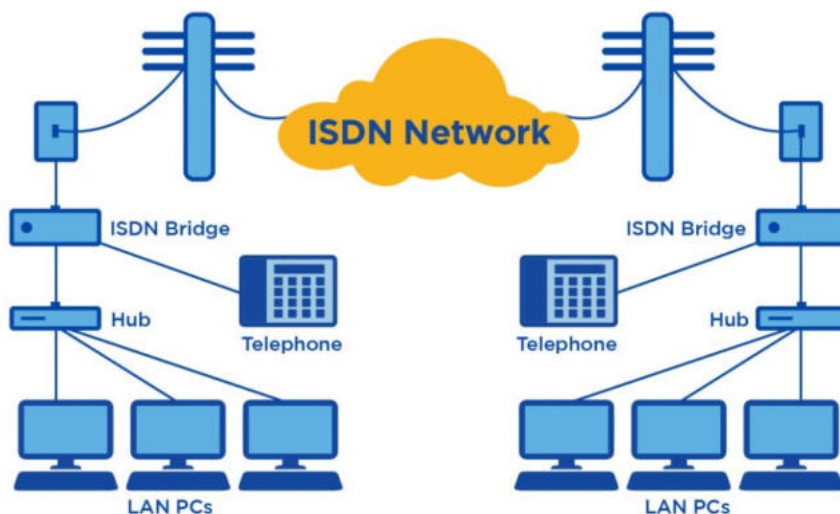
Dial-up Modem

- **DSL Modem:** DSL stands for Digital Subscriber Line. Like dial-up modem, DSL modem also uses telephone lines to transfer digital signals. It provides faster speed than dial-up and supports both data and voice transmission over phone lines.



DSL Modem

- **ISDN Modem:** Integrated Services Digital Network is a digital phone connection that can transmit data, voice and video over a normal telephone line at the same time which was not done before. It is faster and expensive technology. Since ISDN work on digital transmission, it converts analog voice to digital signals before transmission.



Types of Computer Networks

Networks are categorized based on their size, purpose, and the geographical area they cover. They can cover anything from a handful of devices within a single room to millions of devices spread across the entire globe. There are different types of computer networks:

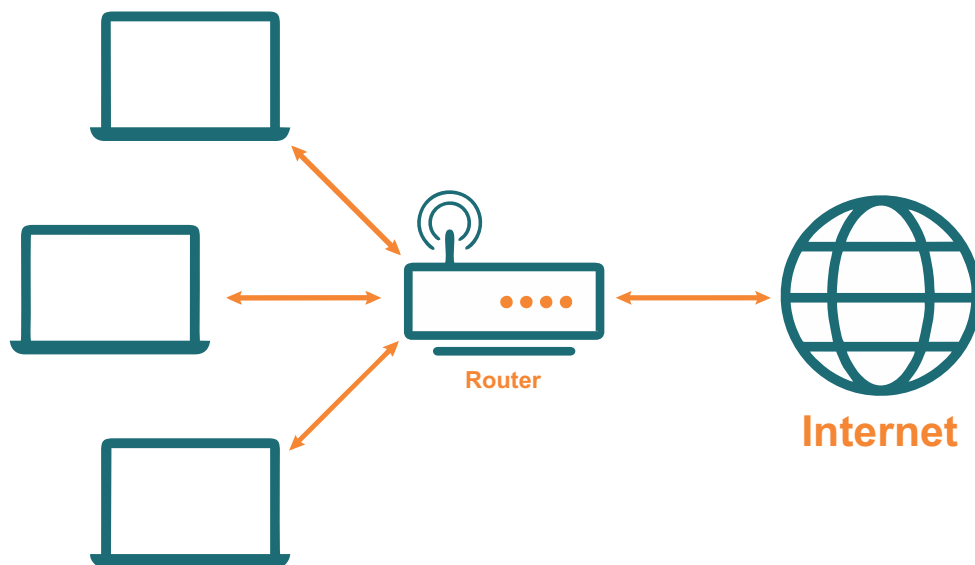
1. Personal Area Network (PAN)

A **Personal Area Network** (PAN) is a small network used for personal devices within a very limited range, such as connecting a phone to a laptop via Bluetooth. It typically covers a few metres around the user.



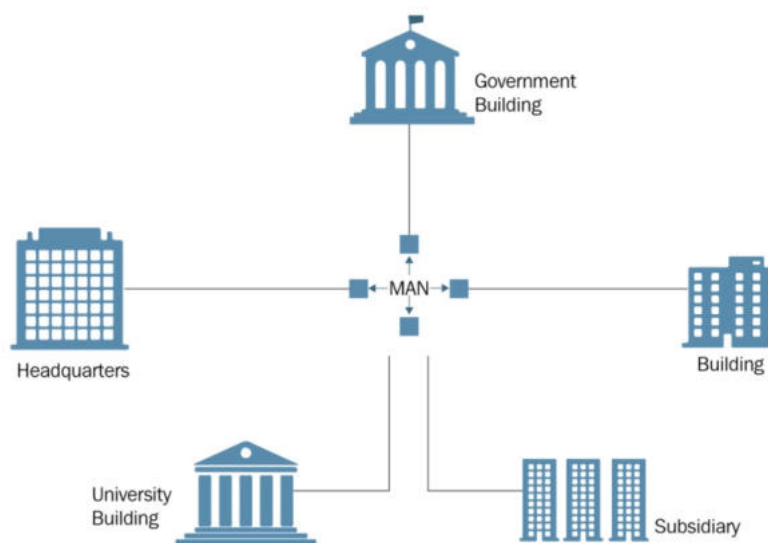
2. Local Area Network (LAN)

A **Local Area Network** (LAN) covers a smaller geographical area, such as a school, office, or home. It connects computers and devices within the same building or campus, enabling them to share resources like printers, servers, and internet connections. Ethernet cables, commonly used in LANs, are physical cables that connect devices such as computers, routers, and switches. These cables transmit data using electrical signals and are typically made of copper wiring.



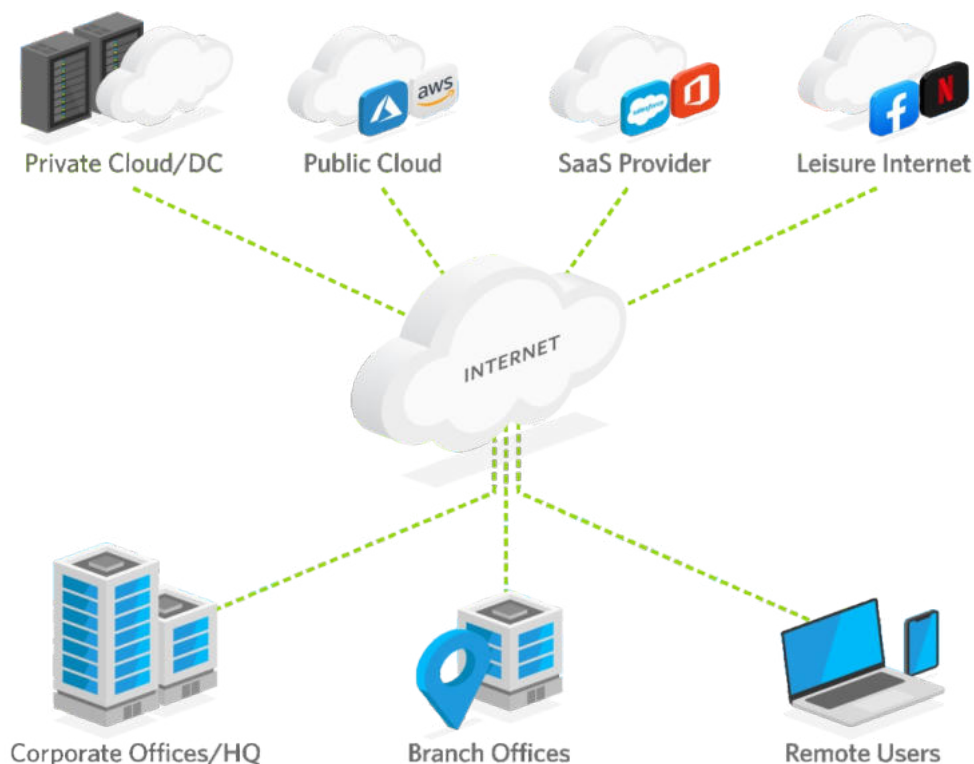
3. Metropolitan Area Network (MAN)

A **Metropolitan Area Network** (MAN) covers a larger area like a city or campus and connects multiple LANs for faster communication over several miles. It's often used by universities, large businesses, or city-wide networks.



4. Wide Area Network (WAN)

A **Wide Area Network** (WAN) covers a large area, such as countries or continents. The **internet** is the most common example of a WAN, connecting people across the world for communication, resource sharing, and data exchange.



Types of Network Connections

Networks use two main types of connections:

1. Wired Connections:

- Wired Connections use physical cables (e.g., Ethernet).

Advantages: High speed, stable connection.

Disadvantages: Limited mobility.

2. Wireless Connections:

- Wireless Connections use signals like Wi-Fi or Bluetooth.

Advantages: High mobility.

Disadvantages: May face interference and slower speeds compared to wired connections.



EXERCISES

01: Choose the correct answer.

1. A network connecting devices in a small area is called:

- ☐ WAN
- ☐ LAN
- ☐ PAN

2. A computer requesting data is a:

- ☐ Client
- ☐ Router
- ☐ Server

3. A router connects:

- ☐ Different networks
- ☐ Devices within a network
- ☐ Only servers

4. The internet is an example of:

- ☐ WAN
- ☐ LAN
- ☐ PAN

5. These connections use cables for communication.

- ☐ Wireless
- ☐ Wired
- ☐ Wi-Fi

6. This connects multiple devices in a network.

- Switch
- Router
- Server

7. Which of the following devices directs data between networks?

- Modem
- Switch
- Router

8. The purpose of a modem is to:

- Connect multiple devices within a network
- Convert digital signals into analog signals
- Direct data between networks

9. This allows devices to communicate wirelessly within a short range.

- LAN
- PAN
- MAN

10. Which type of network connects devices over large geographical areas, such as countries?

- LAN
- WAN
- MAN

02: Fill in the blanks.

1. A _____ provides services or resources to clients.
2. A _____ directs data between networks.
3. _____ connections use signals like Wi-Fi.
4. The largest network in the world is the _____.
5. A _____ connects multiple devices in a network.
6. A _____ is a device that routes data to the right path in a network.

03: Tick True or False for each statement.

	TRUE	FALSE
A LAN is larger than a WAN.	☐	☐
A workstation is used to share resources with the server.	☐	☐
Wired connections are more stable than wireless connections.	☐	☐
A client provides data to servers.	☐	☐
Bluetooth is used in Personal Area Networks.	☐	☐
A modem helps connect devices within a network.	☐	☐

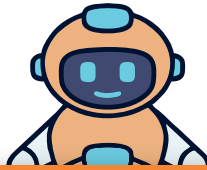
04: Answer the following questions briefly.

1. Define networking.
2. What is the role of a router in a network?
3. List two advantages of wired connections
4. Differentiate between LAN and WAN.
5. What is a client in networking?
6. What does a switch do in a network?

05: Provide detailed answers for the following questions.

1. Explain the components of a network with examples.
2. Describe the differences between wired and wireless connections.
3. What are the different types of networks? Explain with examples.
4. How is networking useful at home and in schools?
5. Why is a switch important in networking?
6. What is the role of a modem in a network?

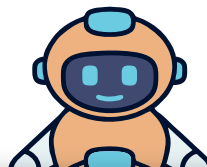
Lab Activity :1



List Network Components:

Identify and list the devices such as clients, servers, and routers in your school or home network.

Lab Activity: 2



Create a LAN:

Use Ethernet cables to connect two computers and share a file between them to understand the basic operation of a Local Area Network (LAN).